Semantic Exploration Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Analysis of Lag-Based Semantic Distance

PENPAL Project

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1 Overview

The semantic exploration outputs now use the **current lag-based structure**:

- `k`: lag
- `distance`: mean semantic distance for that lag
- `agent`: `"interleaved"` for the full exchange sequence, or `author_1` / `author_2` for speaker-specific trajectories

The processed filename is still `semantic_exploration_binned.parquet`, but the current generator computes **lag-based distances**, not the older non-overlapping window-centroid bins.

This notebook therefore no longer assumes old turn-level exploration tables. Instead it analyses:

1. **Interleaved sequence exploration** across conditions
2. **Speaker-specific exploration** using recovered role metadata

Important comparison note:

- **Slot 1 / Slot 2** is the directly comparable cross-condition axis.
- **Human / AI** in Human-AI and **Starter-side / Non-starter-side** in same-type conditions are only used as within-condition descriptive views.

Primary cross-condition object. Per story, we compute the trapezoidal area under each slot's mean distance(k) curve over the intersection of k values present for both slots, and take the unsigned difference:

$$A_{\text{semexp}}(i) \;=\; \left|\,\text{AUC}_{\text{slot}=1,\,i} - \text{AUC}_{\text{slot}=2,\,i}\,\right|.$$

This is label-invariant and comparable across conditions. The lag-binned LMM slot contrast and the log-distance slope view are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded semantic exploration rows by condition and agent view

condition	agent	n_stories	n_rows	min_k	max_k
Human-AI	author_1	97	730	1	8
Human-AI	author_2	97	730	1	8
Human-AI	interleaved	97	970	1	10
Human-Human	author_1	36	288	1	8
Human-Human	author_2	36	288	1	8
Human-Human	interleaved	36	360	1	10
AI-AI	author_1	38	304	1	8
AI-AI	author_2	38	304	1	8
AI-AI	interleaved	38	380	1	10

3 Preprocessing

```
## Interleaved rows: 1710
```

```
## Speaker rows: 2644
```

4 Primary: Per-Story AUC Asymmetry

Table 2: Stories with per-story AUC computable in both slots

condition	n_stories
Human-AI	97
Human-Human	36
AI-AI	38

Table 3: Per-story |AUC(slot 1) - AUC(slot 2)| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	97	0.229	0.174	0.195
Human-Human	36	0.210	0.169	0.173
AI-AI	38	0.214	0.163	0.180

```
## One-way ANOVA: A_semexp ~ condition
##              Df Sum Sq Mean Sq F value Pr(>F)
```

```
## condition     2  0.012 0.00613   0.211   0.81
## Residuals   168  4.884 0.02907
##
## Kruskal-Wallis (robust): A_semexp ~ condition
##
##  Kruskal-Wallis rank sum test
##
## data:  A_semexp by condition
## Kruskal-Wallis chi-squared = 0.45786, df = 2, p-value = 0.7954
##
##
## Pairwise Wilcoxon (BH-adjusted):
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  df_auc_stories$A_semexp and df_auc_stories$condition
##
##             Human-AI Human-Human
## Human-Human 0.9      -
## AI-AI       0.9      0.9
##
## P value adjustment method: BH
```

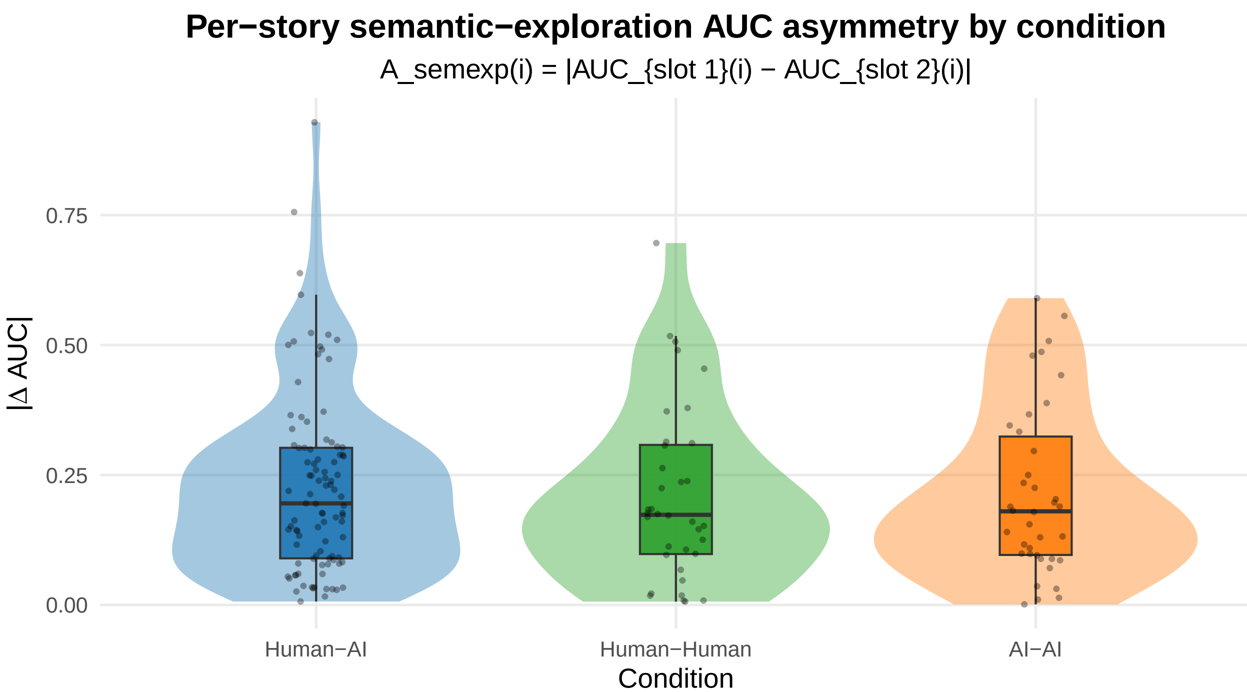


Table 4: Secondary: per-story |log(distance) ~ log(k) slope diff| by condition

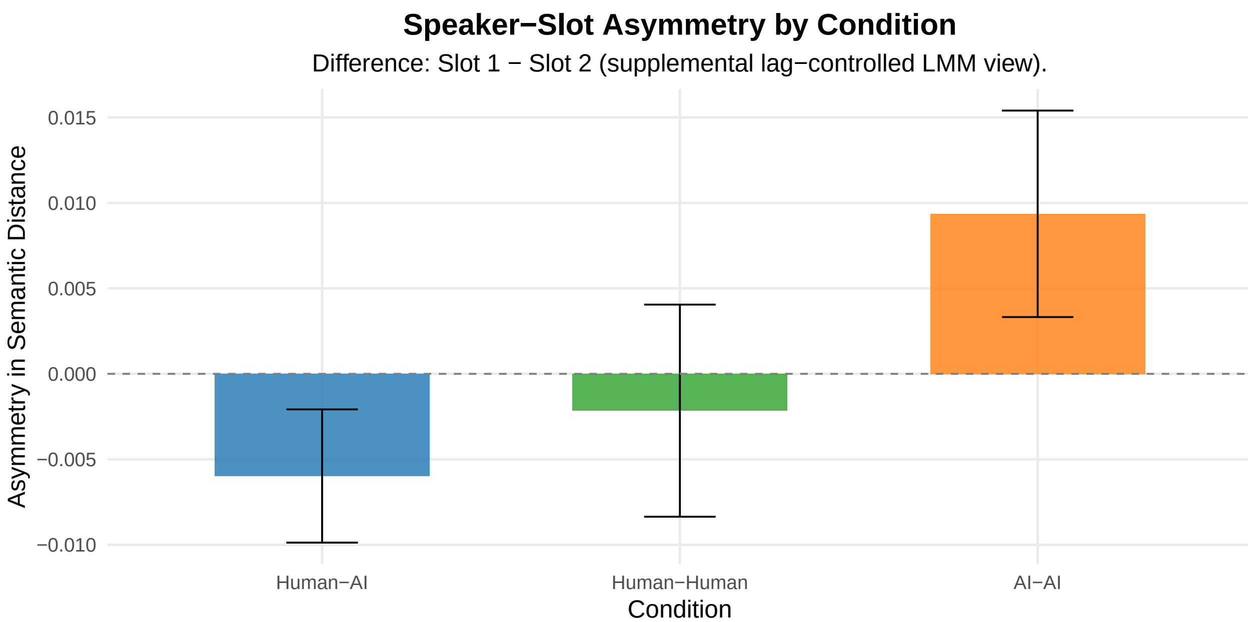
condition	n	mean_A	median_A
Human-AI	97	0.092	0.079
Human-Human	36	0.095	0.083
AI-AI	38	0.079	0.068

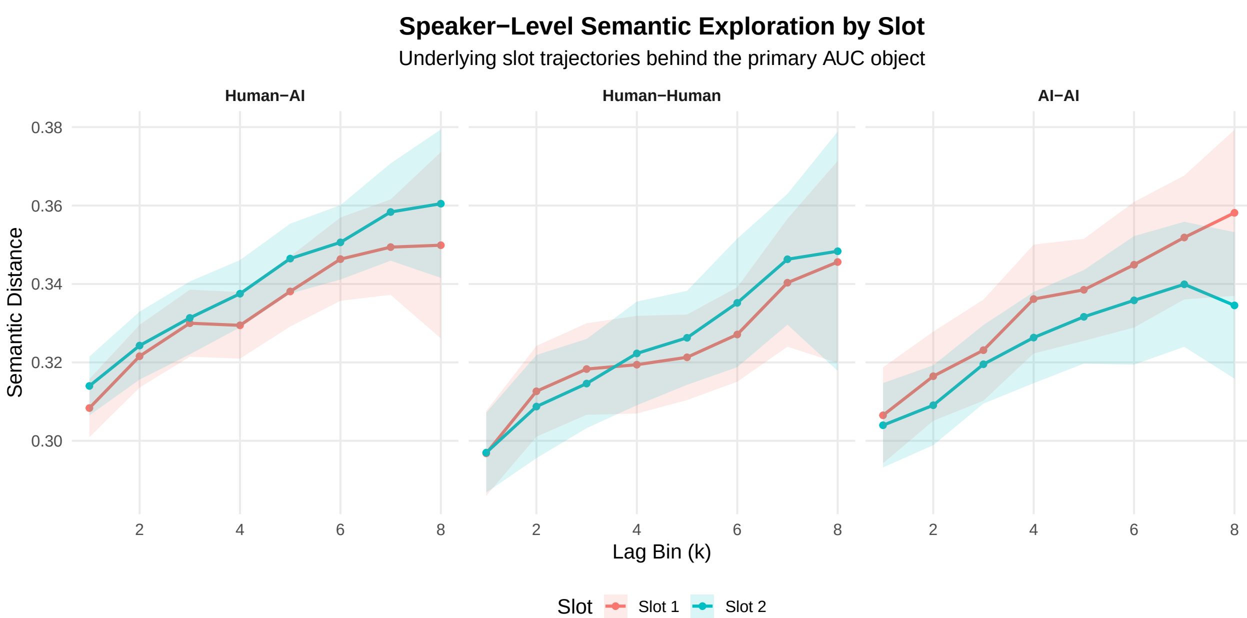
5 Supplemental: Lag-Binned Slot Contrast

```
## Speaker-slot asymmetry model:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance ~ k + slot * condition + (1 | conversation_id)
##    Data: df_speaker
##
## REML criterion at convergence: -9327.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2878 -0.5867 -0.0045  0.5827  5.2251
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0008745 0.02957
##  Residual                    0.0014426 0.03798
## Number of obs: 2644, groups:  conversation_id, 171
##
## Fixed effects:
##                                   Estimate Std. Error         df t value
## (Intercept)                      3.057e-01  3.606e-03  2.834e+02  84.772
## k                                6.421e-03  3.319e-04  2.473e+03  19.343
## slotSlot 2                       5.975e-03  1.988e-03  2.469e+03   3.005
## conditionHuman-Human            -1.186e-02  6.348e-03  2.007e+02  -1.868
## conditionAI-AI                  -8.872e-05  6.226e-03  2.008e+02  -0.014
## slotSlot 2:conditionHuman-Human -3.822e-03  3.738e-03  2.469e+03  -1.023
## slotSlot 2:conditionAI-AI       -1.534e-02  3.666e-03  2.469e+03  -4.183
##                                 Pr(>|t|)
## (Intercept)                      < 2e-16 ***
## k                                < 2e-16 ***
## slotSlot 2                       0.00268 **
## conditionHuman-Human             0.06316 .
## conditionAI-AI                   0.98864
## slotSlot 2:conditionHuman-Human  0.30664
## slotSlot 2:conditionAI-AI       2.98e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      sltSl2 cndH-H cAI-AI sS2:H-
## k           -0.393
## slotSlot 2  -0.276  0.000
## cndtnHmn-Hm -0.476 -0.012  0.157
## condtnAI-AI -0.485 -0.013  0.160  0.278
## sltS2:cnH-H  0.147  0.000 -0.532 -0.294 -0.085
## sltS2:AI-AI  0.149  0.000 -0.542 -0.085 -0.294  0.288
```

Table 5: Speaker-slot asymmetry in semantic exploration by condition

condition	difference	se	t	p
Human-AI	-0.006	0.002	-3.005	0.003
Human-Human	-0.002	0.003	-0.680	0.496
AI-AI	0.009	0.003	3.039	0.002

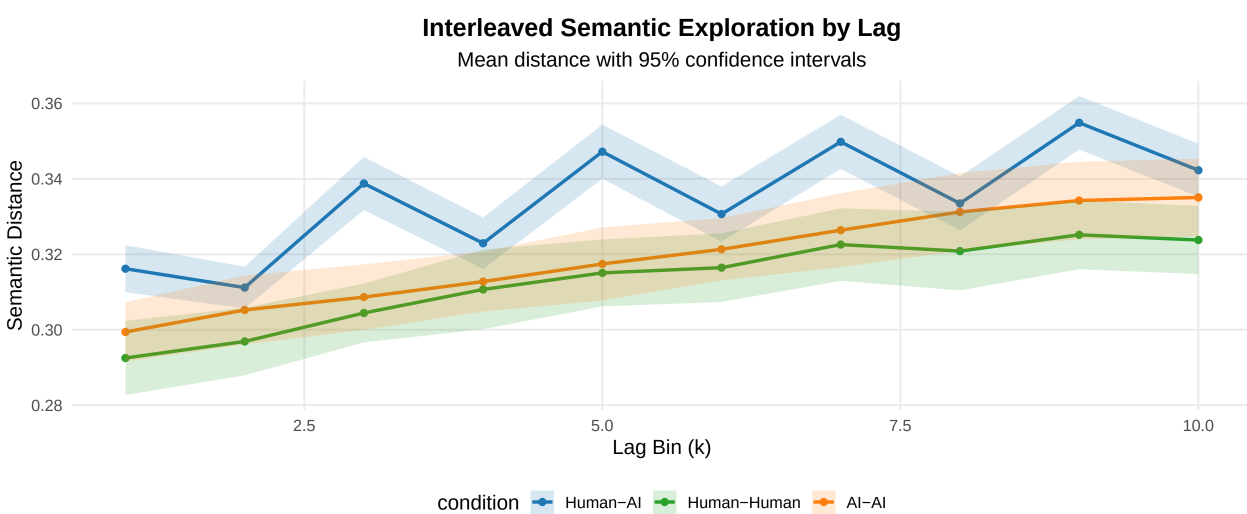




6 Supplemental: General Condition-Level Comparison

Table 6: Interleaved semantic distance by lag and condition

condition	k	mean_distance	sd_distance	n
Human-AI	1	0.316	0.031	97
Human-AI	2	0.311	0.028	97
Human-AI	3	0.339	0.035	97
Human-AI	4	0.323	0.034	97
Human-AI	5	0.347	0.036	97
Human-AI	6	0.331	0.037	97
Human-AI	7	0.350	0.036	97
Human-AI	8	0.333	0.036	97
Human-AI	9	0.355	0.036	97
Human-AI	10	0.342	0.035	97
Human-Human	1	0.293	0.030	36
Human-Human	2	0.297	0.028	36
Human-Human	3	0.304	0.024	36
Human-Human	4	0.311	0.032	36
Human-Human	5	0.315	0.027	36
Human-Human	6	0.316	0.028	36
Human-Human	7	0.323	0.029	36
Human-Human	8	0.321	0.032	36
Human-Human	9	0.325	0.028	36
Human-Human	10	0.324	0.028	36
AI-AI	1	0.299	0.025	38
AI-AI	2	0.305	0.029	38
AI-AI	3	0.309	0.027	38
AI-AI	4	0.313	0.025	38
AI-AI	5	0.317	0.031	38
AI-AI	6	0.321	0.026	38
AI-AI	7	0.326	0.031	38
AI-AI	8	0.331	0.032	38
AI-AI	9	0.334	0.032	38
AI-AI	10	0.335	0.032	38




```
## Interleaved model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

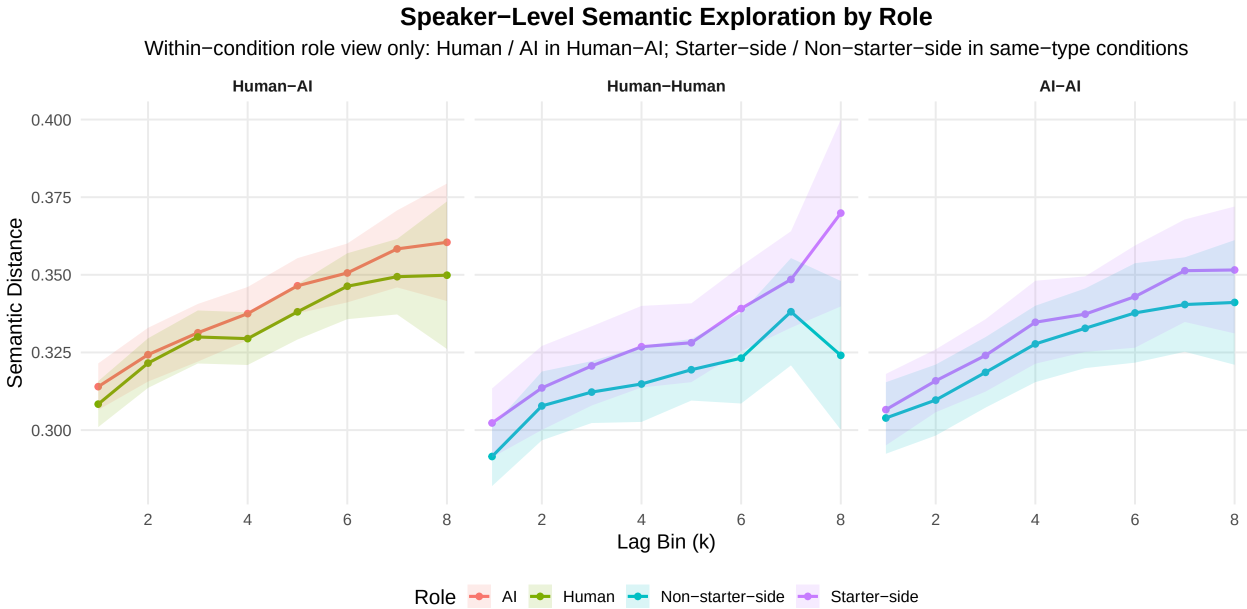
```
## lmerModLmerTest]
## Formula: distance ~ k * condition + (1 | conversation_id)
##    Data: df_interleaved
##
## REML criterion at convergence: -8436.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5704 -0.5955  0.0224  0.6190  3.0341
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0008042 0.02836
##  Residual                    0.0002900 0.01703
## Number of obs: 1710, groups:  conversation_id, 171
##
## Fixed effects:
##                          Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)             3.154e-01  3.112e-03  2.132e+02 101.357  < 2e-16 ***
## k                       3.506e-03  1.904e-04  1.536e+03  18.419  < 2e-16 ***
## conditionHuman-Human   -2.256e-02  5.982e-03  2.132e+02  -3.771 0.000211 ***
## conditionAI-AI         -1.900e-02  5.866e-03  2.132e+02  -3.239 0.001392 **
## k:conditionHuman-Human  1.208e-04  3.659e-04  1.536e+03   0.330 0.741435
## k:conditionAI-AI        6.254e-04  3.588e-04  1.536e+03   1.743 0.081566 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      cndH-H cAI-AI k:cH-H
## k           -0.336
## cndtnHmn-Hm -0.520  0.175
## condtnAI-AI -0.531  0.178  0.276
## k:cndtnHm-H  0.175 -0.520 -0.336 -0.093
## k:cndtAI-AI  0.178 -0.531 -0.093 -0.336  0.276

##
## Condition contrasts (interleaved view):

##  contrast                    estimate      SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  0.02189 0.00563 168   3.887  0.0004
##  (Human-AI) - (AI-AI)        0.01556 0.00552 168   2.817  0.0163
##  (Human-Human) - (AI-AI)    -0.00633 0.00671 168  -0.944  1.0000
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

7 Supplemental: Role-Level View

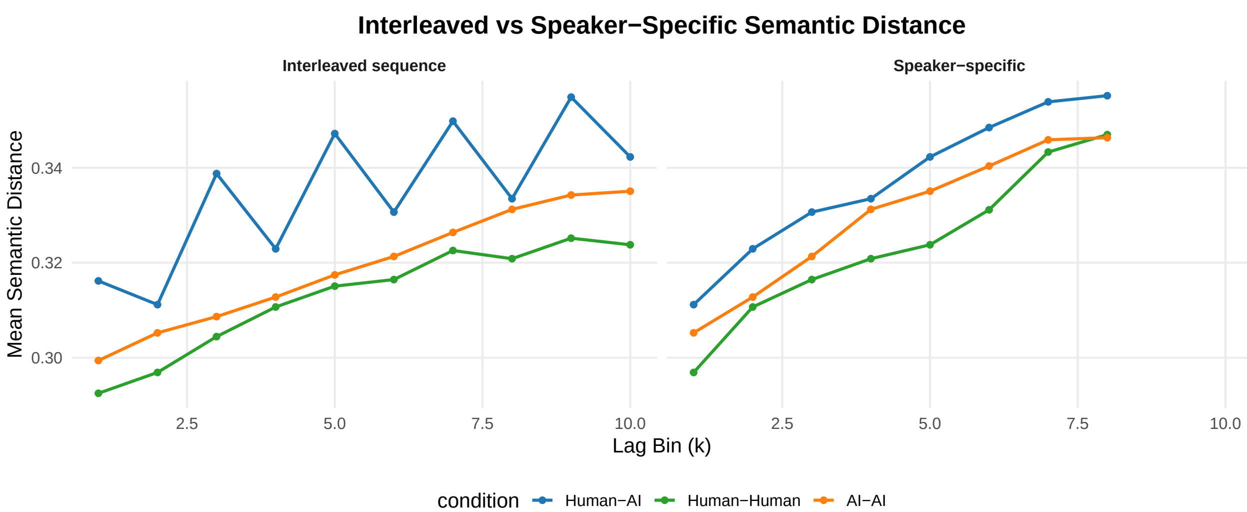
Table 7: Speaker-level semantic distance by condition, role, and lag

condition	condition_role	k	mean_distance	n
Human-AI	AI	1	0.314	97

condition	condition_role	k	mean_distance	n
Human-AI	AI	2	0.324	97
Human-AI	AI	3	0.331	97
Human-AI	AI	4	0.338	97
Human-AI	AI	5	0.346	97
Human-AI	AI	6	0.351	97
Human-AI	AI	7	0.358	96
Human-AI	AI	8	0.360	52
Human-AI	Human	1	0.308	97
Human-AI	Human	2	0.322	97
Human-AI	Human	3	0.330	97
Human-AI	Human	4	0.329	97
Human-AI	Human	5	0.338	97
Human-AI	Human	6	0.346	97
Human-AI	Human	7	0.349	96
Human-AI	Human	8	0.350	52
Human-Human	Non-starter-side	1	0.291	36
Human-Human	Non-starter-side	2	0.308	36
Human-Human	Non-starter-side	3	0.312	36
Human-Human	Non-starter-side	4	0.315	36
Human-Human	Non-starter-side	5	0.319	36
Human-Human	Non-starter-side	6	0.323	36
Human-Human	Non-starter-side	7	0.338	36
Human-Human	Non-starter-side	8	0.324	36
Human-Human	Starter-side	1	0.302	36
Human-Human	Starter-side	2	0.314	36
Human-Human	Starter-side	3	0.321	36
Human-Human	Starter-side	4	0.327	36
Human-Human	Starter-side	5	0.328	36
Human-Human	Starter-side	6	0.339	36
Human-Human	Starter-side	7	0.349	36
Human-Human	Starter-side	8	0.370	36
AI-AI	Non-starter-side	1	0.304	38
AI-AI	Non-starter-side	2	0.310	38
AI-AI	Non-starter-side	3	0.319	38
AI-AI	Non-starter-side	4	0.328	38
AI-AI	Non-starter-side	5	0.333	38
AI-AI	Non-starter-side	6	0.338	38
AI-AI	Non-starter-side	7	0.340	38
AI-AI	Non-starter-side	8	0.341	38
AI-AI	Starter-side	1	0.307	38
AI-AI	Starter-side	2	0.316	38
AI-AI	Starter-side	3	0.324	38
AI-AI	Starter-side	4	0.335	38
AI-AI	Starter-side	5	0.337	38
AI-AI	Starter-side	6	0.343	38
AI-AI	Starter-side	7	0.351	38
AI-AI	Starter-side	8	0.352	38



8 Supplemental: Interleaved vs Speaker View



9 Summary

Primary cross-condition result: per-story $A_{\text{semexp}} = |\text{AUC}_{\text{slot 1}} - \text{AUC}_{\text{slot 2}}|$ from trapezoidal integration of the distance(k) curve over the intersected k range. Tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. A secondary log-distance-slope asymmetry is reported for dynamic-shape continuity with the paper.

Supplemental views retained:

- lag-controlled slot-contrast LMM (`distance ~ k + slot * condition`)
- interleaved sequence model and condition descriptives
- within-condition role curves (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type)
- interleaved vs speaker-specific side-by-side view

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized AUC asymmetry across all conditions)
- Human-AI decomposition (`speaker_type \u00d7 speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-26 22:35:34.565779